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Coalgebraic notions of simulation and bisimulation

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Let F be an endofunctor on a category $\mathcal{C}$. An *F-coalgebra* is a pair (X, α) consisting of an object X, the *carrier*, and a morphism $\alpha\colon X \to FX$, the *structure*. F-coalgebras model a wide range of state-based systems. For $\mathcal{C} = \mathbf{Set}$, if $F = \mathcal{P}$ is the powerset functor, then F-coalgebras are Kripke frames. If $FX = \mathcal{P}(A \times X)$ for a fixed set A, then F-coalgebras are labelled transition systems. If $FX = (\mathcal{D}X)^A$, where $\mathcal{D}$ is the distribution functor, then F-coalgebras are labeled Markov chains [3].

A bisimulation between two state-based systems is a relation that relates states with the same behaviour. A simulation is a relation that relates a state s of one system to a state t of the other only if t can do whatever s does, and possibly more. Every bisimulation is a simulation, but the converse fails in general.

Simulation and bisimulation can be defined abstractly for a pair of F-coalgebras. We refer to these definitions as *coalgebraic (bi)simulation*. Several notions of coalgebraic bisimulation have been studied, and their relationship is understood [1]. The situation for coalgebraic simulation is different. Simulation is asymmetric, and its abstract definition requires additional structure. The relationship between the different notions of coalgebraic simulation is not clarified in the literature, and neither is their relationship to the notions of coalgebraic bisimulation.

Current research on coalgebraic simulation is based on *relators* [2]. A relator for an endofunctor F on **Set** is a monotone map $\tilde{F}$ that sends every relation $r\colon X \rightarrowtail\!\!\!\!\!\!\rightarrow Y$ to a relation $\tilde{F}r\colon FX \rightarrowtail\!\!\!\!\!\!\rightarrow FY$. An endofunctor on **Set** in general has several relators. A standard example is the Barr extension $\bar{F}$ of F. A relation r is an *$\tilde{F}$-simulation* from an F-coalgebra (X, α) to an F-coalgebra (Y, β) if $r \subseteq \beta^{\mathsf{op}} \cdot \tilde{F}r \cdot \alpha$. A relator $\tilde{F}$ is *symmetric* if $\tilde{F}(r^{\mathsf{op}}) = (\tilde{F}r)^{\mathsf{op}}$ for every relation r. For a symmetric relator $\tilde{F}$, $\tilde{F}$-simulations are called *$\tilde{F}$-bisimulations*. The Barr extension $\bar{F}$ is symmetric.

Relators yield different notions of simulation and bisimulation for a single endofunctor on **Set**, but the theory of relators is restricted to **Set**. Another notion of simulation, *Aczel–Mendler simulation* [4], is defined over a general underlying category. It is a relaxation of the well-known *Aczel–Mendler bisimulation* [3]. The relationship between Aczel–Mendler simulation and relator-based simulation is not yet clear. A further goal is a notion of relator for endofunctors on an arbitrary category, giving several notions of simulation and bisimulation for one endofunctor. As a step in this direction, we propose *Hermida–Jacobs simulation*. It relaxes *Hermida–Jacobs bisimulation* [1] in the same way that Aczel–Mendler simulation relaxes Aczel–Mendler bisimulation. Its generality is limited: it generalises only the Barr extension, which is one particular relator for each functor.

In this talk, we review coalgebraic notions of simulation and bisimulation and the connections between them. We identify open questions about these connections and present the answers we have obtained so far.

Keywords: Category theory; Coalgebra; (Bi)simulations
Subject area: Coalgebra

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